

01/04/25

Java Assignment Operator

1. int I = 6; $\rightarrow$ 6

2. int J = 7; $\rightarrow$ 7

3. int I = J; $\rightarrow$ 7

J value copied to I

The Important thing to understand with assignment is the fact that, This is different from Mathematics

* When I say I is equal to J [It does not mean Both are equal]

* In programming language it telling I is to J is [Copy the value of J into I]

* The Assignment Operator is used to copy the value of one ~~into~~ variable into another.

* In an assignment The Value on the left-hand side should always be a variable.
You cannot say.

$10 = 5$ X Error

* An assignment can be done with expressions as well.

$i = 10; j = 20$

i) $i = j * 2 \rightarrow 50$

ii) $i = i * 2 \rightarrow 30$

iii) $i = i - i \rightarrow 0$

iv) $i = i + 1 \rightarrow 1$

v) $i = i + 1 \rightarrow 2$

Java Assignment Operator - puzzles on Increment And Decrement And compound

1. int $A = 5; \rightarrow 5$

2. int $A = A + 1 \rightarrow 6$

3. int $A++ \rightarrow 6$

~~For~~ This method $A++ [6+1]$ but its output give 6.

4. int $A \rightarrow 7$

but in memory it stored as 7

5. int $A++ \rightarrow 7$ [it print old value but it's increment the Value]

6. int $A \rightarrow 8$

7. `int A--` $\Rightarrow$ 8
 8. `int A` $\Rightarrow$ 7 :- } The Value is decremented.

Other method.

9. `int ++A` $\rightarrow$ 8
~~1+1~~ } First incremented with Value
 Then printed output

<code>++A</code>	} $\rightarrow$	Used to increment Increment Value
<code>A++</code>		
<code>--A</code>	} $\rightarrow$	Used to decrement Value
<code>A--</code>		

$\rightarrow$ Compound assignment operators

* The Compound assignment operators help you to simplify your code

[So, if I am doing `i = i + 2`; Let's say instead of that I can say `i += 2`]

1. `int i = 10;`

`i += 2` $\rightarrow$ 12
~~10 + 2~~

2. `i -= 2` $\rightarrow$ 10
~~12 - 2~~

3. `i *= 2` $\rightarrow$ 20
~~10 * 2~~

→ Java Conditionals and if Statement

Unfortunately in java the syntax ^{for is} of ~~if~~ is not this simple.

for loop syntax

```
for (initialisation; condition; update)
    Statement;
```

[* If I want to repeat this statement 10 times, Then I would need to do something called iteration,

* Then I would need to check a condition

* Then I would need to define and update statement

* And after that is when can execute the statement

* The important part of the for loop is called a Condition. [A condition either true or false

* If condition is true, the for loop continues execution

* If condition is false, the for loop ~~can~~ stop execution

→ What is a condition?

example:- 1. int i = 10

$i < 5 \rightarrow \text{false}$

i has the value 10. is i less than 5 no so it is false
This is called condition

$i = 10$

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ii) $i > 5 \rightarrow \text{true}$ ^{yes}
is 1 Greater Than 5 ~~as~~ it is true

iii) $i \leq 5 \rightarrow \text{false}$
is 1 Less Than equal to 5 no it is false

iv) $i \geq 5 \rightarrow \text{true}$
is 1 Greater Than equal to 5 yes it is true

(*) The Important thing to understand is that a condition returns a boolean Value

$\rightarrow$ The Syntax If

If (Condition)
Statement;

$i = 10$

i) If ($i < 5$)

System.out.println("i is less Than 5");

$\downarrow$

Nothing Printed.

because i Value is 10. it is a false Condition
That why it is not printed

2) int $i = 3$; if ($i < 3$) $\rightarrow$ Condition

System.out.println("i is less Than 5");
 $\nwarrow$ Statement

$\downarrow$

i is less Than 5

The Output is Printed because The condition is true

Exercises

1. Have four Variables a, b, c, d and create an if statement to print if $a+b$ is greater than $c+d$.
2. Have three angles of a triangle stored in three Variables $angle1, angle2, angle3$. Create an if statement to state if the three angles can form a triangle. Hint: $angle1 + angle2 + angle3 = 180$
3. Have a Variable Called $number$. Create an if statement to find if it is even number. Hint: $\%$ operator

Solution

1. $int\ a=6; int\ b=2; int\ c=3; int\ d=1;$

if ($a+b > c+d$)

System.out.println("a+b is greater than (c+d)");



"a+b is greater than c+d"

2. $int\ angle1=90; int\ angle2=90; int\ angle3=90;$

if ($angle1 + angle2 + angle3 == 180$)

System.out.println("Valid Triangle");



Valid Triangle

3. int number = 10;

if (number % 2 == 0)

System.out.println("It is even number")



It is even number